

from harddisk to the RAM & it stick to RAM even after its execution,

enable on sticky bit on Repeatedly executable programs will speed up the execution as it avoid Reading the file from hard disk Every time its executes

SGID:- when enabled on executable files the program Runs with group privileges

when SGID is enabled on directories, all the child directories and files inherits the group privileges.

~~Notes~~

05/01/19

Access Control list:- Access Control list provides an additional more flexible permissions for a file system or files or directory. Acls allow fine-grained permissions to be allocated to a file named users or named groups as well as users and groups identified by a UID or GUID, can be granted permissions in addition to standard file owner, group owner and other file permissions. file owner can set Acl's on individual files or directories, the file system need to be enabled with ACL option.

file system mount option for Acl's:- The file system needs to be mounted with ACL Support enabled. xfs (Extended file system) file systems have a builtin ACL support. ext4 file systems created on RHEL7 have the ACL option enabled by default. Earlier versions may need the ACL option included with the mount Request or set on the superblock.

Eg:- vi /etc/fstab

dev name	dir	filetype	defaults, acl	0	0
/dev/sdc1	/oracle	ext4	<u>defaults, acl</u>	0	0

(or)

mount -o remount, acl. /mountpoint

The remount option on the above command will Reload newly given mount options without the need of unmount.

viewing and interpreting ACL permissions:- 'ls -l' Command will only prints minimal ACL settings.

Ex:- ls -l xyz.txt

```
-rw-rw-r-- + 1 root root 34 Nov 10 21:43 xyz.txt
```

The '+' symbol at the end of the 10th character permission indicates that there are ACL settings associated with this file.

viewing file ACLs:- To display ACL settings on a file use

getfacl file/directory.

Ex:- getfacl u.txt

#file: u.txt

user::rw-

#owner: root

group::r--

#group: root

other::r--

#flags: s-t

In the above file 1st four lines are commented, indicates file name, owner, group & flags respectively. flags refers to file attributes such as setuid, setgid and sticky bit.

user entries:-

1. The 1st user entry refers to the file owner permission whereas root is having r&w permission,

2. The 2nd user entry refers the name dbadm associated with this file.

user::rw-

named user: dbadm:rw-

group entries:- The group permission having Read access and named group oracle is having read and write access.

mask entry:- mask setting indicates the maximum possible permissions for all named users, group owner & named groups but not the file grower.

Setting up ACLs:- use setup ACL for setfacl command to add, modify or Remove standard ACLs on files and directories

Syntax:- setfacl -m u:username:rwX g:groupname:rwX file/directory

Setfacl -R -m u:username directory
 ↓
 Recursive g: group
 o: other

Setfacl -d -m u:username:rw- directory
 ↓
 default

usually default acls are applied on directory.

Example:

~~touch~~ mysample.txt

ls -l

Setfacl -m u:sample1:rw- mysample.txt

getfacl mysample.txt

Setfacl -m u:sample2:rw-, g:hr:r-x, g:fin:--- mysample.txt

getfacl mysample.txt

getfacl

Note:- chmod command has no effect on any group permissions for files with ACL settings, but it updates the ACL mask.

To add or modify other ACL:- ACL's other and standard other permissions are Equivalent, so using chmod on other permissions is Equivalent to using 'setfacl' or other options.

Eg:- setfacl -m o:rw- xyz.txt

(or) chmod o=rw- xyz.txt

We can add multiple entries via same command by using comma separation for each entry.

Eg:- setfacl -m u:sample2:rw-, g:hr:r-x, g:fin:---, o:--- mysample.txt

The above command will set named user sample2 with rw- & named group hr with r-x, & named group fin with no permissions and others also with no permissions.

The ACL mask:- The ACL mask defines the maximum permissions that can be granted to named users, group owner, and named groups. It doesn't restrict the permission of the file owner or other users. All files and directories that implement ACL's will have an ACL mask.

The mask will be calculated and added automatically if it not explicitly set. The mask also get inherited from the parent directory if a default mask is set.

default ACL entries:- The default ACL can be applied on owner, named user, group owner and a named group. The file owner will get the file permissions as narrated below. The default owner the named user will obtain read & write on a permissions for files created on the directory & Read, write & execution permission for directory created on the directory.

Ex:- mkdir sap

```
cd sap
ls -ld sap
touch x.txt
ls -ld
ls -ld sap
getfacl sap
cd sap
touch x.txt
ls -ld
getfacl x.txt
mkdir subdir
ls -ld
getfacl subdir/
```

Removing all ACL Setting for file or directory:-

Syntax:- setfacl -b file/dir
↓

Removes all ACL's onto of given file/dir.

Removing default ACL:-

Syntax:- setfacl -k file/dir.

Removing specific ACL Entry:-

Setting a ACL's of one file to another:-

Syntax:- `getfacl x.txt > acl.out`

`setfacl --set-file=acl.out y.txt`

`getfacl y.txt`

`touch z.txt`

`setfacl --set-file=acl.out z.txt`

(or)

`getfacl x.txt | setfacl --setfile= - m.txt`

modifying or changing ACL's with Set - option:-

you can use `--set` (or) `--set-file` options to set ACL's of a file/directory. The output from `getfacl` can be used as a input to `setfacl` command

Eg:- `getfacl x.txt | setfacl --setfile: - m.txt`

The `--set-file` option accepts input from a file (or) stdin, and the '-' (hyphen) specifies the use of stdin. on the above case m.txt will have same ACL's as x.txt.

storing ACL's onto a file:-

Syntax:- `getfacl filename > acloutfile`

apply the above created file on some other file as below

`setfacl --set-file=file=acl.out file/dir.`

deleting an ACL:- deleting specific ACL

`setfacl -x u:username:permission filename`

To remove all ACL's `setfacl -b file/dir`

1. create group called 'sodor'

create a user called James

create a group called a) bakerstreet b) scotlandyard.

create users lestrade, gregson, Jones as part of scotlandyard.

create users holmes, watson as part of bakerstreet.

a) add named ACL's to the steamies directory and all of its contents

i) grant the sodor group read, write & conditional execute permissions.

ii) update steamies directory, denying the user james from the sodor group any access.

b) add named ACL's as default ACL's to support future file & directory additions.

c) add default access rule for the sodor group, grant read, write & execute permissions on the steamies directory.

d) and add default access rule for the user james. deny all access to the steamies directory.

2. The Baker street detective agency is setting up a collaborative share directory and copy some of its contents to hold case files which members of the bakerstreet group will have read & write permission on.

The lead detective, Sherlock Holmes, has decided that members of the scotlandyard group should also be able to read & write to the shared directory. However, Jones thinks that inspector Peter Jones (a member of the scotlandyard group) is an imbecile, and as such Jones should have his access to the directory restricted to read only.

Mrs. Hudson has limited linux skills and was only able to create

groupadd Sodor.

groupadd bakerstreet

groupadd Scotlandyard

useradd James, useradd Jones, useradd Lustrade, useradd gregson,

useradd Holmes, useradd Watson.

mkdir

groupmems -a Lustrade -g Scotlandyard.

" " gregson -g "

" " Jones -g "

mkdir steamies

getfacl Sodor

groupmems -a James -g Sodor

setfacl -d -m g:Sodor:rw steamies

setfacl -d g:Sodor:rw steamies

setfacl -d -m g:bakerstreet:rw share

" " " g:Scotlandyard:rw share.

setfacl -d -m u:Jones:r-- g:Scotlandyard share

setfacl -d -m u:Jones:r--, g:bakerstreet:rw-, g:Scotlandyard:rw- share

getfacl share

cd share

mkdir cases

cd cases

touch adventure.txt, territory.txt

ls -l

getfacl

passwd listtrade Set Redhat passwd

" gregson

"

"

"

" Jones

"

"

"

" Holmes

"

"

"

" walson

"

"

"

ls -l

check getfacl shares cases/

06/01/2019

su Command:- Switch user also known as substitute user.

Syntax:- su username

(or)

su - username

(or)

su - (or) su

su -c command username

su -c whoami dbadm.

when '-' is used along with su command (su - dbadm) make the shell, clears all environment variables except for term (terminal), initializes home, shell, user, logname and path.

when -c is used (su -c command username) passes a single command to the shell.

observation:-

x=100

echo \$x

export x

100

echo \$x

su - dbadm

output 100

undefined variable.

su dbadm

echo \$x

output 100

The best practice is to use `su - username` where certain environment variables will be applied.

Observation:- `#whoami`

`root`

`#who am i`

`root pts/2`

`# su dbadm`

`# whoami`

`dbadm`

`# who am i`

`root pts/2`

`# exit`

`# su - dbadm`

`# whoami`

`dbadm`

`# who am i`

`root pts/2`

difference between `su` and `su -` ?

mounting a dvd:

Syntax: `mount -o ro /dev/cdrom /mountpoint`

↑
device name

↓ ↓
option readonly

(or)

`mount -o ro -t iso9660 /dev/cdrom /mountpoint`

↓
type

To unmount dvd `umount /mnt`

Sudo Command :- (ebaidik language) Sudo su -

what are the permissions of /etc/sudoers

-r--r-----

what and all users scenarios you have faced in your project?

userlimits:- ulimit

- Configuration file : /etc/security/limits.conf

The /etc/security/limits.conf file contains Stanzas that specify the process Resource limits for each user (or) a group. each line describes a limit for a user in the form <domain> <type> <item> <value>

where <domain> can be 1. username.

2. groupname, @group syntax (/. can be also used).

3. The wildcard * for default entry.

<type> can be 1. Soft -

2. Hard - maximum limit for the soft limit.

<item> can be 1. core - limits the core file size (KB) - kilobyte

2. data - max data size in KB

3. fsize - max file size in KB.

4. nofile - max no. of opened file descriptors

5. rss - max resident set size

6. cpu - max cpu time in minutes

7. nproc - max no. of processes etc.

<value> can be value Refers the associated value to limit the user.

use a value of '-1' is used to set a resource to unlimited.

ulimit -a

ulimit -f 1024

ulimit -f

1024

ulimit -f unlimited

df -i

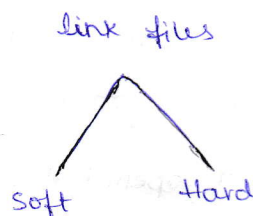
ls -li

stat mfile

1. what is file system structure?
2. what is Superblock?
3. what is inode & what it consists of?
4. how will you print inode info? `ls -li`
5. can you change no. of inodes of a filesystem?
6. how will you print a file inodes of filesystem.
7. what happens if I run `mkfs` cmd?
8. what are datablocks & why we need to have datablocks?
9. why only 4kb for datablocks?
10. what `ls -l` Command do?

07/01/19

link files:- In mfile link/mfile $\rightarrow$ In file target



hard link cannot be allowed for hardlink directory.

It can allow only files but in directory it consists of two links (..) & (.)

(..) Refers to parent directory and

(.) Refers to current directory.

`ls -lia`

* How will you manipulate your inode numbers?

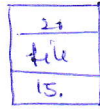
* How many ^{max} links you can create for a file? $2^{32} - 1$ if it is 32 bit file system

Have you ever tried No.

`ln -f` if the file already exists it will be overwritten.

Soft link:- Syntax:- `ln -s link3 slink1`

`ls -li` = `slink` → `link3`



directory link is only possible with softlink.

hard link can't be done across file system.

Hard link:- Syntax:- `ln sourcefile targetfile` → new file.

already existed
↑

Eg:- `ln x.txt y.txt`

more than one file can be referenced by a same inode number and these files said to be hard linked each other.

directory as a file is just an array of filenames. where a directory is created it initially populates filenames (`.` and `..` special files).

the `.` is a file that maps with the current directory.

`..` maps to the parent directory.

Softlink:-

Syntax:- `ln -s sourcefile targetfile`

A softlink is a file which contains the name of other file.

→ list of files opened on filesystem.

Syntax:- `lsdf /datafs01` → `lsdf` (mountpoint name)

Scenario:- when admin is trying to unmount the filesystem we will get device is busy the reason for the same is programs/applications/database/users might be using the filesystem and hence the filesystem is busy. to solve the problem you need to examine which processes or which user using the file system

below are the commands - are using same.

1) `fuser` - `fuser` identifies processes used by `dirname`

fuser -u -k /fsol



kill all process used by given file system

(You need to have proper approvals to kill programs/daemons etc under the given filesystem)

2. lsof → list opened files

Syntax: lsof /fsol

lsof prints the open files processes along with their pids which can be ~~manually~~ killed. You unmount along the filesystem however as being it is a destruction activity makesure approval from business/client / application owner.

File system consistency check: fsck is used to check filesystem consistency. It cannot check if the filesystem is mounted.

Syntax: fsck /dev/datavq/datavq.

Note: do not run fsck on mounted file system.

e2fsck -n /mnt/pt or /dev/vg/lv



do not correct issue.

Answer: NO

e2fsck -y /mnt/pt or /dev/vg/lv



yes

Filesystem Journaling: A special area on the filesystem is reserved to keep filesystem transactions is known as Journal.

If any abnormal unmount of a filesystem or power outage etc. will cause filesystem corruption. In this case the Journal will be replayed to complete the filesystem transactions.

difference between ext2, ext3, ext4?

ext2	ext3	ext4
- There is no Journal feature available - max file size (individual) = 16GB to 2TB max. FS size = 2TB to 32TB - supported from	- Journal feature available - max. individual file size = 16GB to 2TB But FS size can be ^{2TB to} 32TB - supported from kernel ver: 2.4.15 - Can have 32000 max subdirs	- Journal feature available along with multiblock allocation - max. file size = 16TB max FS size = 1EB (1 Exa byte = 1024PB) - supported from 2.6.19 onwards 64000 max. subdirs - performance Reliability is more compared to ext3

Converting ext2 to ext3:- if you are upgrading /dev/datvq/dlvo1 i.e., mounted on /home, to convert ext2 to ext3 do the following:-

1. umount /dev/datvq/dlvo1 (or) umount /oracle
2. tune2fs -j /dev/datvq/dlvo1 creating Journal. This will add a Journal.
3. use below Command to check which filesystem.

file -sL /dev/datvq/dlvo1.

and again mount the device

note:- you really don't need to unmount while converting ext2 to ext3, it can happen dynamically. Converting ext3 to ext4:- when file system is mounted (dynamically). However it is recommended to doing the conversion of file line.

Converting ext3 to ext4:-

1. umount /oracle.

* differences phases of ~~fsck~~ fsck?

when we run fsck below are the phases:-

1. pass 1 - checking inodes, blocks & sizes

pass 2 - " directory structure

pass 3 - " " connectivity

pass 4 - " reference counts

pass 5 - " group summary information.

Process priority:-

ps -e shows all the entire process.

ps -e | more shows page by page

Every process is identified by unique id starts with 'P'.

ps

PID	TTY	TIME CMD
↓	↓	↓
processid	teletype	cpu time (sys+user)
3572	pts/0	00:00:00 bash.

processname/command.

? means not started by any terminal.

Zombie or defunct process Represented by 'Z'. it is a process that has completed its execution but still have entry on process table.

F S UID PID PPID C PRI MI ADDR SZ WCHAN STIME TTY TIME CMD

F - flag

S - state S, R, P, T, Z (Common States)

Attempting to killing zombie process will not succeed. you might be Restart System once again however zombie process may not Refer on the above.

on the above ps -ef Refers to

F - flag. flag usually an integer number whereas '1' Refers to the state that the process has been forked but didn't execute. and '4' is used for superuser privileges

UID - process owner's user ID or name.

PID - process ID

C - CPU utilization.

PPID - Parent process ID.

PRI - priority. The priority of a process can be identified by its priority (PRI) and nice value (NI).

There are exactly 40 different levels of niceness a process can have.

The nice levels of a process ranges from -20 to 19. By default processes inherit nice level from its parent which is usually '0'. Higher nice values indicates less priority, while lower nice levels indicates a higher priority.

only Root user can increase the priority of a process and the owner of the process can lower the priority.

Launching process with different priority:-

Syntax:- `nice -n <nicelvl> /path/to/command.`

where nicelvl can be -20 to 19.

unprivileged users are only allowed to set a positive nicelvl (0 to 19) and only root can set a negative nicelvl.

renice: This is the command is applied on the Job that are already running to alter the priority of a Job.

`renice -n <nicelvl> path to command. pid`

Eg:- `nice -n -10 /home/tejaswini/shellscript#prg10.sh`

`renice -n -20 4351.`

`renice -n 19 4351.`

find Command:- It is used to search for files on a directory hierarchy.

Syntax:- `find /path -name file/dirname -print.`

The path is an absolute path where we suppose to search file or directory.

`find /home/tejaswini -name prg1.sh -print.`

Eg:- `#find / -name passwd -print`

`(etc/passwd`

The above Command attempt find passwd file on the system and print its absolute path if it finds.

2. find a file prgl.sh on /root

```
find /root -name prgl.sh -print
```

3. To print all the files on the system.

```
find / -print
```

4. To print all the files on /home/tejaswini

```
find /home/tejaswini -print
```

5. You can use shell metacharacters on find Command. Below examples shows how to find all partition devices under /dev/sdb.

```
find /dev -name sdb? -print.
```

6. find all scsi devices.

```
find /dev -name sd* -print.
```

7. To find filenames owned by user tejaswini

```
find / -user tejaswini -print
```

8. To search files by ignoring Case sensitiveness

```
find / -iname passwd -print.
```

9. To find all files that are owned by given group

```
find / -group oracle -print
```

10. To find all files or Dir owned by user id 1000

```
find / -uid 1000 -print
```

11. To find files owned by particular user and group

```
find / -user dbadmin -group oracle -print
```

12. finding files based on permissions

```
find / -perm 755 -print
```

```
find /home/shellscripts -perm 644 -print.
```

13. Searching files based on size attributes

a) find files that are 100MB on size

```
find / -size 10M -print
```

b) find files that are greater than 10MB on size

```
find / -size +10M -print
```

14. Search files based on its modification timestamp:

a) find files that are modified 120min ago

```
find / -mmin 120 -print
```

b) find files that are modified greater than 120min

```
find / -mmin +120 -print
```

c) find files that are modified less than 120min

```
find / -mmin -120 -print
```

15. Search files based on its type:

a) find all directories

```
find / -type d -print
```

b) find all Regular files

```
find / -type f -print
```

c) find all link files

```
find / -type l -print
```

d) find all block special files

```
find / -type b -print
```

16. search a file on given device

```
find / -name xyz.txt -xdev -print
```

17. find files with inode num

```
find / -inum 35968 -print
```

-xdev is given if search will happen on that device only

```
find /root/scripts -name prog1*.sh -exec mv {} /tmp/16-11;
```

```
find /tmp/programs -inum 651768 -exec rm -f {} \;
```

ls -l

It Removes particular file with the inode number.

03/01/19

gzip:- It is used to Compress a file.

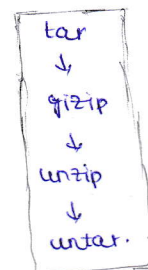
Ex:- gzip filename This Command creates filename.gz Compressed file.

gunzip:- It is used to uncompress gzip format file

Ex:- gunzip filename.gz

How to see Particular file is there on tar?

```
tar -xvf dir.tar | grep
```



tar Command (Tape Archive):-

Archiving means Safeguarding ^d data

TAR:- tape archive

Command to archive file/directories

```
tar -cvf tarfilename files to archive
    -xvf
    -tvf
```

Where c - create tar file

v - verbose (list while doing tar)

f - tar filename

x - extract file tar file

Examples:- `tar -cvf ash.tar /tarfs`

1041 `tar -xvf ash.tar`

1046 `gzip ash.tar`

1051 `tar -cvf /tmp/etc.tar/etc`

1053 `cd /tmp/`

1055 `tar -tvf etc.tar`

1062 `du -h etc.tar`

1065 `gzip etc.tar`

1067 `gunzip etc.tar.gz`

Note:- we can do tar and compression by using `-Z` flag

```
tar -cvzf aa.tar.gz /myfiledir
```

To untar in single step

```
tar -xvzf filename.tar.gz
```

Compressing a file:- `gzip filename`

filename extensions are `.gz`

uncompressing a file:- ~~gzip~~ `gunzip filename.`